ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- **Transient analysis of 1st and 2nd systems**
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 8b: AC Circuit Analysis

- ✓ **Transient analysis of 1st and 2nd systems**

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Introduction to Transient Analysis

Transient Analysis

- Transient analysis studies how a circuit or a network behaves over time after a sudden change, such as when a switch opens or closes.
- When a circuit contains energy storage elements (inductors $L$and capacitors C), it does not respond instantly to such changes. Instead, it undergoes a **transition period** before reaching a steady-state condition.
- Transient analysis focuses on understanding and characterising this transition period in detail.

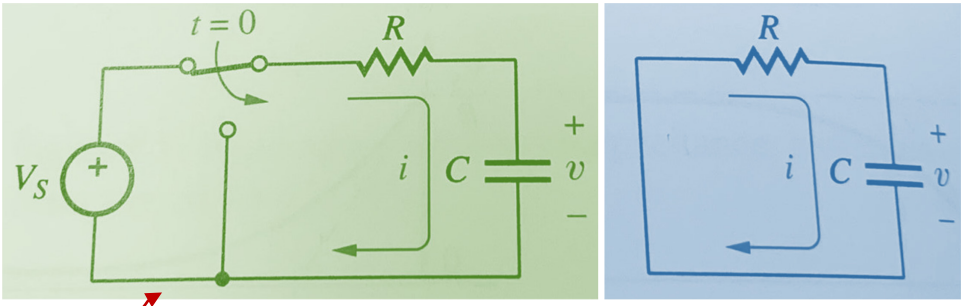
Two types of circuits:

- **First-order circuits:** These contain only one energy storage element (either an inductor or a capacitor). Such circuits are described by a first-order differential equation.
- **Second-order circuits:** These contain both an inductor and a capacitor. As a result, the circuit behaviour is described by a second-order differential equation.

Source-free circuits: Natural response

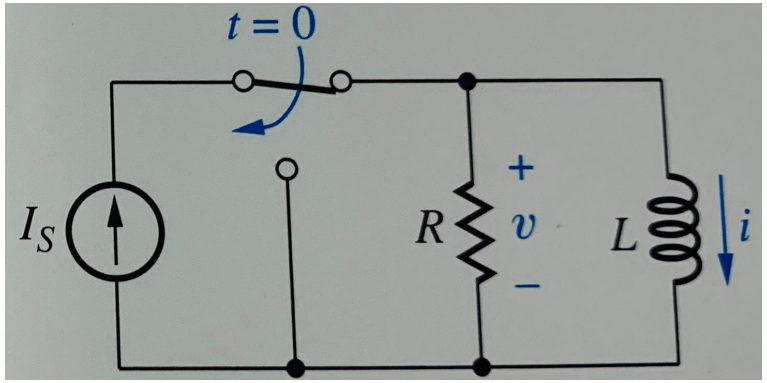
✓ A **source-free RC or RL circuit** occurs when its independent source is suddenly switched off. The energy previously stored in the capacitor or inductor is then released through the resistor.

- The **natural response** of a circuit refers to the behaviour of voltages and currents after the source is switched off at $t = 0$ (See Figure (a) on the right)
- For a **source-free RC circuit**, the natural response involves finding the **voltage across the capacitor** C for $t > 0$ by solving the governing differential equation (see Figure (b) on the right)
- For a **source-free RL circuit**, the natural response involves finding the **current through the inductor** L for $t > 0$ by solving the governing differential equation.



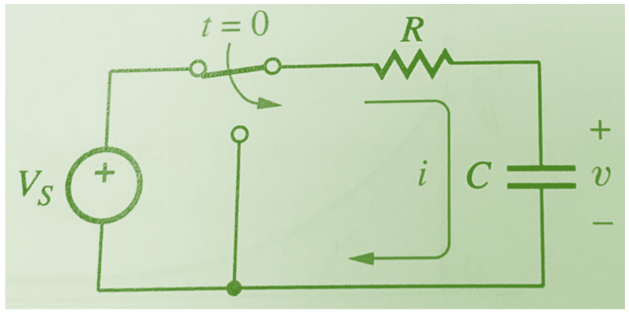
(a) (b)

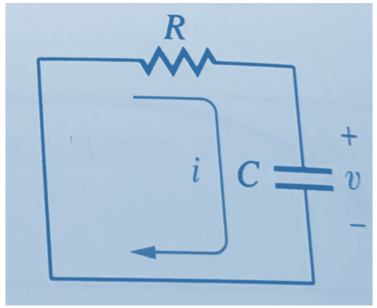
- Suppose that after the capacitor in Figure (a) has fully charged to V_s ,the switch is flipped back down at $t = 0$.
- With the source removed, the circuit becomes a source-free circuit, as shown in Figure (b).



- If the switch is left in the up position for a sufficiently long time, all of $I_s$will eventually flow through the inductor L, and none through the resistor R.
- If the switch is then flipped down at $t = 0$, the circuit becomes a source-free circuit

Natural Response of a RC circuit



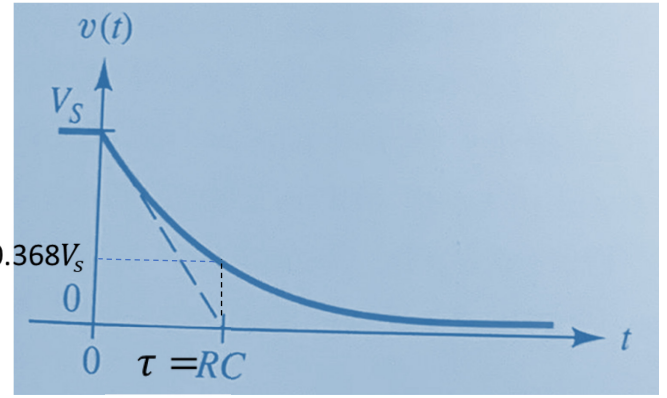


$$i = C\frac{d\,v}{d\,t}$$

- Suppose that after the capacitor in Figure (a) has fully charged to V_s ,the switch is flipped back down at $t = 0$.
- If the switch is then flipped down at $t = 0$, the circuit becomes a source-free circuit

The time constant $\tau$of a circuit is the time required for the response to decay to $1/e$ **(about 36.8%) of its initial value**.

- The response decays **faster** when $\tau$is small
- The response decays **slower** when $\tau$is large



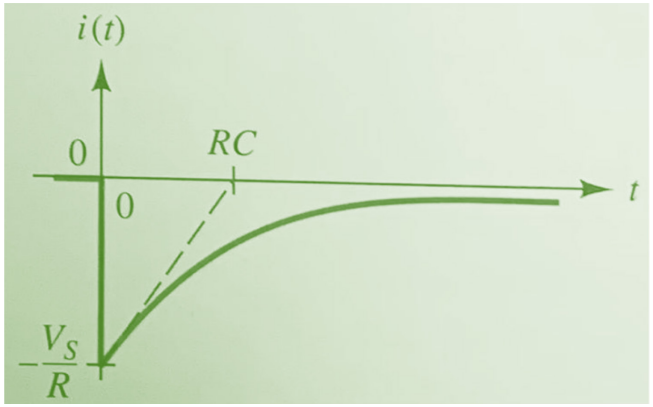
- Let us assume the capacitor $C$is initially fully charged to a voltage V_s at $t = 0$. { at t = 0, $v(0) = V_s$}
- In order to find the natural response, we need to find the voltage across the capacitor C for t > 0 by solving the differential equation,

$$\frac{v}{R} + C\frac{dv}{dt} = 0 \Rightarrow \frac{dv}{v} = -\frac{dt}{CR}$$

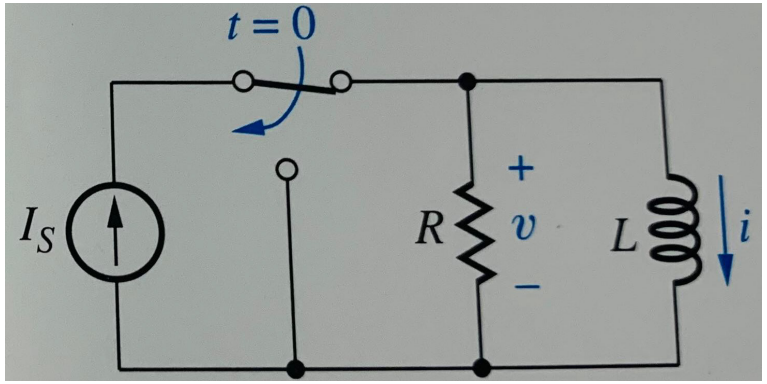
By integrating both sides of the above equation, we obtain

$$\ln v = -\frac{t}{RC} + \ln A \quad \text{(A is a constant)} \Rightarrow v(t) = Ae^{-t/RC} \Rightarrow \boldsymbol{v(t) = V_s\, e^{-t/RC}}$$

The current is readily found as $i(t) = C\frac{dv}{dt} = -\frac{\boldsymbol{V_s}}{R}\boldsymbol{e^{-t/RC}}$ {Time constant: $\tau = RC$}



Natural Response of a RL circuit



- If the switch is left in the up position for a sufficiently long time, all of I_s will eventually flow through the inductor L, and none through the resistor R. The final stored energy in the inductor is $w_L = (\frac{1}{2})\, LI_s^2$

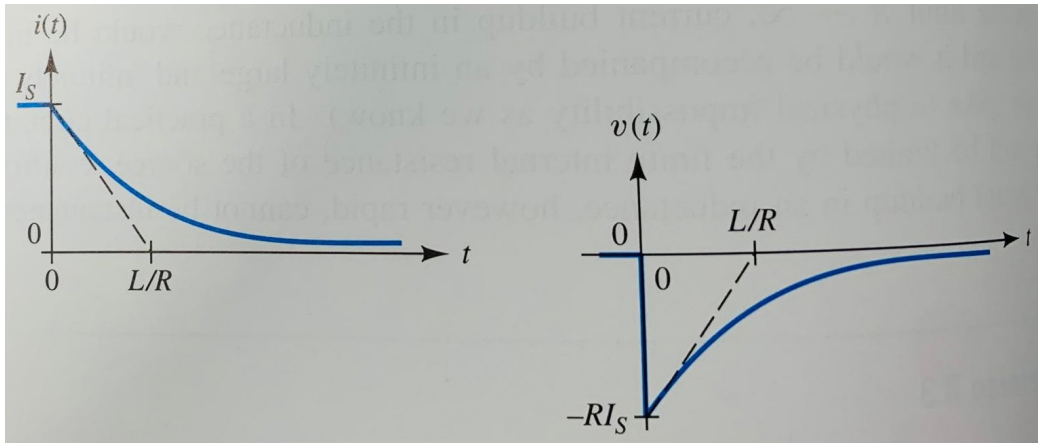
- Suppose now the switch is flipped back down at $t = 0, i(0) = I_s$ and $v(o) = -R\, I_s$

$$i(t)R + L\frac{di(t)}{dt} = 0 \Rightarrow i(t) = I_s e^{-t/(\frac{L}{R})}$$

$$\text{The voltage is found as } \boldsymbol{v(t)} = L\frac{di(t)}{dt} = -RI_s e^{-t/(\frac{L}{R})}$$

{Time constant: $\tau = R/L$}

- The responses $i(t)$ and $v(t)$ are shown below:

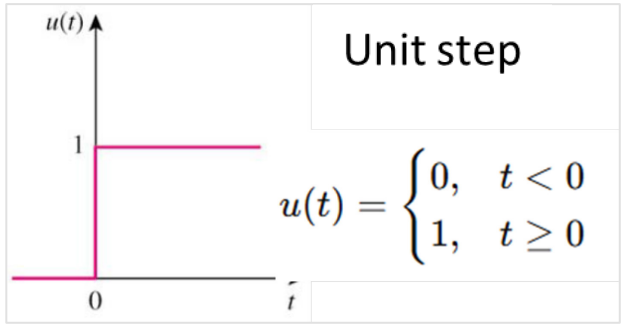


Forced Response

- So far, we have examined networks where a voltage or current source is suddenly removed. As a result, the voltages and currents in the circuit decay to zero, which is known as the natural response of the circuit.

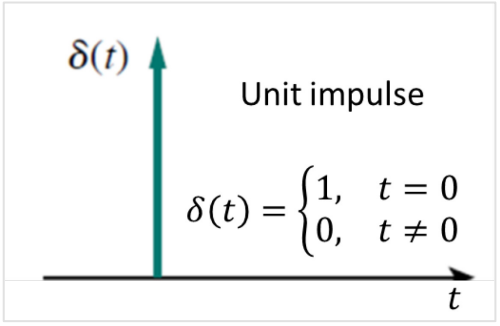
- Next, we will study the response of first-order circuits when an independent DC voltage or current source is suddenly applied. This sudden application forces the voltages and currents in the circuit to change.

- Most widely used functions are:



Unit step

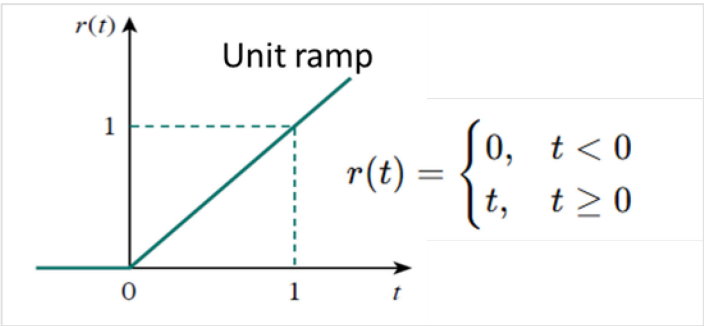
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



Unit impulse

$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

Derivative of unit step function



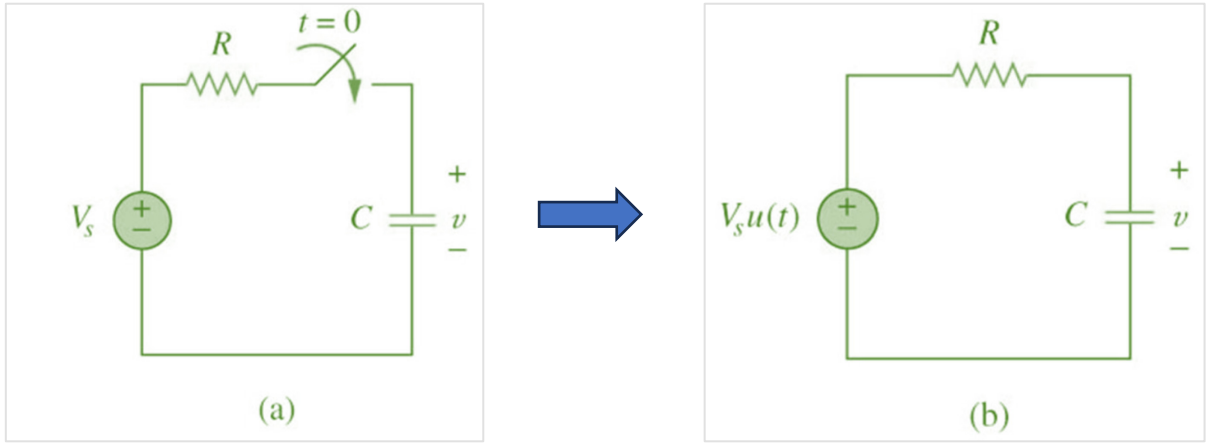
Unit ramp

$$r(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}$$

Integral of unit step function

The Step-Response of a RC Circuit

- When the DC source is suddenly applied, it can be modelled as step function and response of the circuit is known as step-response.



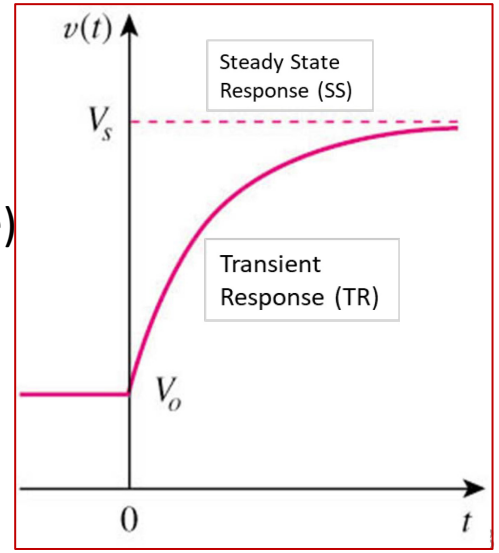
Initial Condition: $v(0) = V_o$

Applying KCL: $c\frac{dv(t)}{dt} + \frac{v(t) - V_s u(t)}{R} = 0$

$$\frac{dv(t)}{dt} + \frac{v(t)}{CR} = \frac{V_s u(t)}{CR}$$

$$v(t) = V_s + (V_o - V_s)e^{-t/RC} \quad t > 0$$

- $v(t) = V_s + (V_o - V_s)e^{-t/RC} \quad t > 0$
  - V_s: Steady State Response (SS)
  - $(V_o - V_s)e^{-t/RC}$: Transient Response (TR)

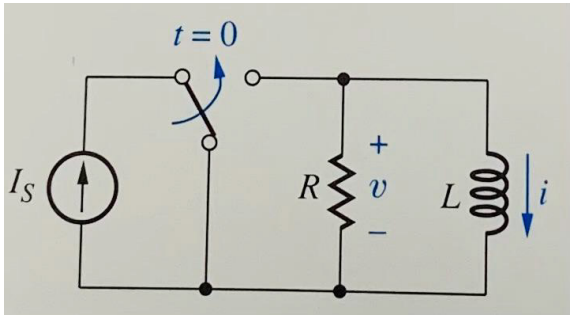


- Complete Response = Natural response (Stored Energy) + Force Response (Independent source)

$$v(t) = V_0 e^{-t/RC} + V_s(1 - e^{-t/RC}) \quad t > 0$$

- The complete Solution is: $v(t) = v_n(t) + v_f(t)$

Forced Response of a RL circuit – Step Response

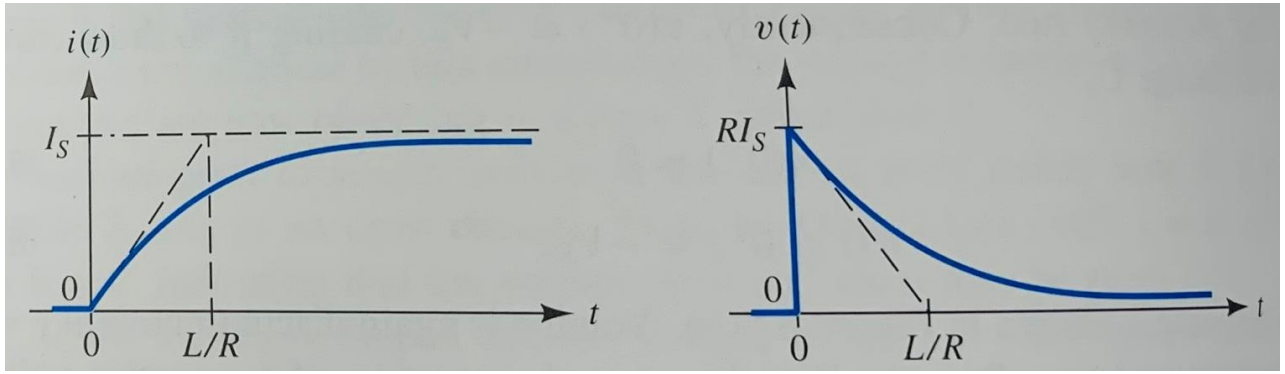


- In the circuit, assume the inductor has zero initial stored energy, so that $i(0) = 0$.
- Flipping the switch connects I_s to the RL pair, causing current and hence energy buildup in the inductor.

$$v = L\frac{d\,i}{d\,t}$$

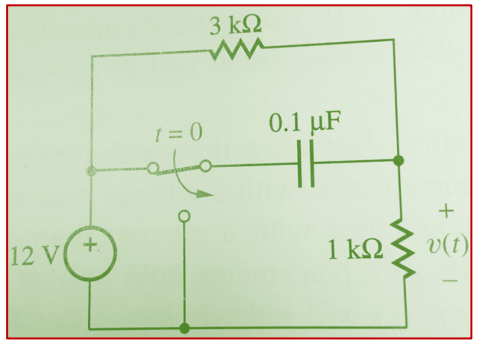
- We have at $t = 0, i(0) = 0\}$
- For $t > 0$, $I_s = \frac{v(t)}{R} + i(t) \Rightarrow i(t) + \frac{L}{R}\frac{di(t)}{dt} = I_s \Rightarrow i(t) = I_s\, e^{-t/(\frac{L}{R})} \Rightarrow v(t) = RI_s e^{-t/(\frac{L}{R})}$

  {Time constant: $\tau = L/R$}
- The responses $i(t)$ and $v(t)$ are shown below:



Example 1

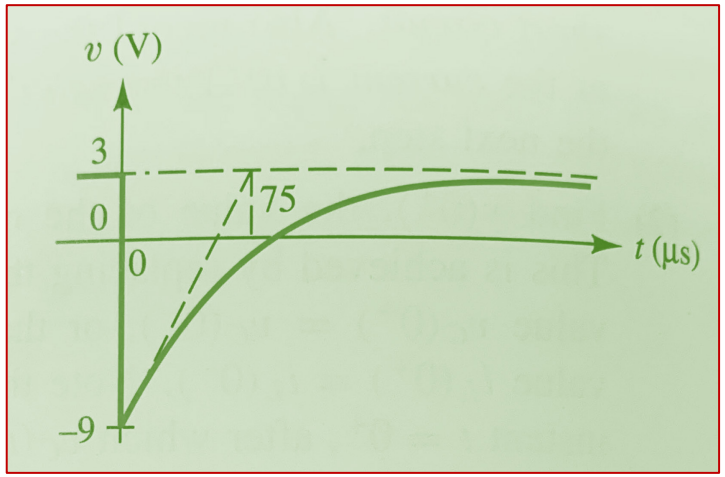
- Sketch and label $v(t)$ for $t > 0$ for the circuit shown below and show $v(t) = 3 - 12\, e^{-t/(75\mu s)}$

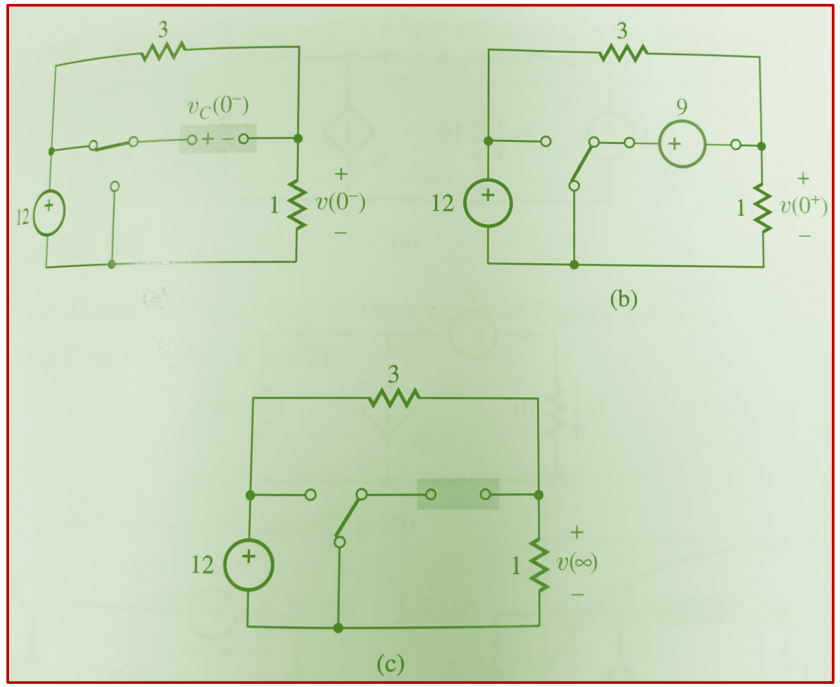


Model Answer

- We need to find $v(0^-), v(0^+), v(\infty)$ and τ *and* the sequence is illustrated on the right .
- $v(0^-) = \left[\frac{1}{3+1}\right] 12 = 3V$; $v_c(0^-) = 12 - 3 = 9V$; $v(0^+) = -9V$; $v(\infty) = 3V$
- R_{eq}= 3//1= 750 ohm; $\tau = RC = 750 \,.\, 10^{-7} = 75\ \mu s$

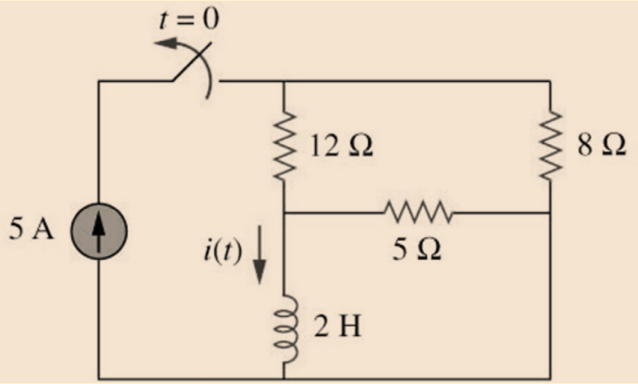
$\therefore v(t) = 3 - 12\, e^{-t/(75\mu s)}$





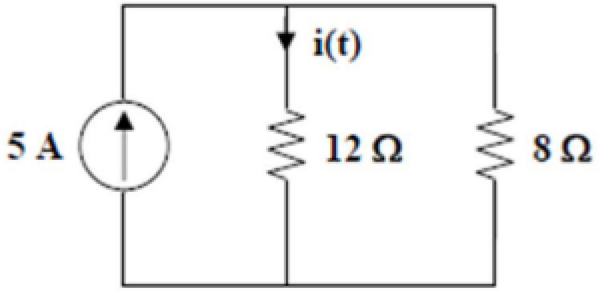
Example 2

- For the circuit, find $i(t)$ for $t > 0$.



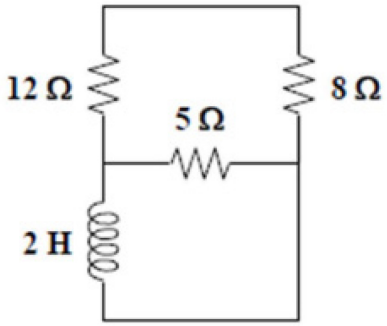
1. Find $i(0) = I_0$. Note that when switch was closed, inductor is shorted for the DC source.
2. Find R_{eq}
3. Find τ
4. When the switch opens, circuit is source free. Find $i(t) = I_0 e^{-t/\tau}$

- For $t < 0$, the eqivalent circuit is shown below:



$$i(0) = \frac{8}{8+12}(5) = 2\ \text{A}$$

- For $t > 0$, the current source is cutt off and the RL circuit is shown below:



$$R_{eq} = (12+8) \parallel 5 = 20 \parallel 5 = 4\ \Omega, \qquad \tau = \frac{L}{R_{eq}} = \frac{2}{4} = 0.5$$

- $i(t) = I_0 e^{-t/\tau} = 2e^{-2t}\ A \quad t > 0$

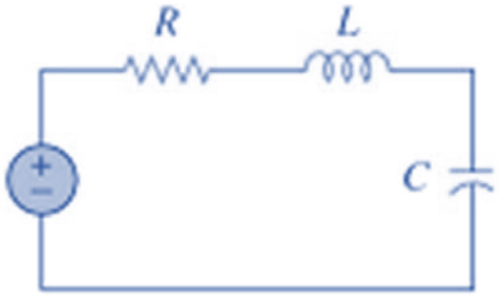
Source – Free Series RLC Circuits

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

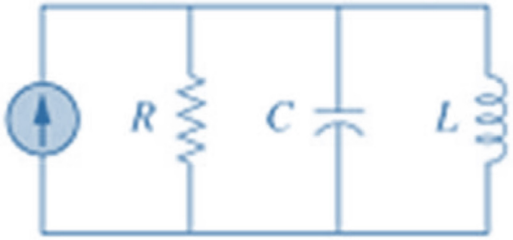
Term 3, 2026

What is a 2nd order circuit?

- A second-order circuit is characterized by a second-order differential equation. It consists of resistors and the equivalent of two energy storage elements



RLC Series

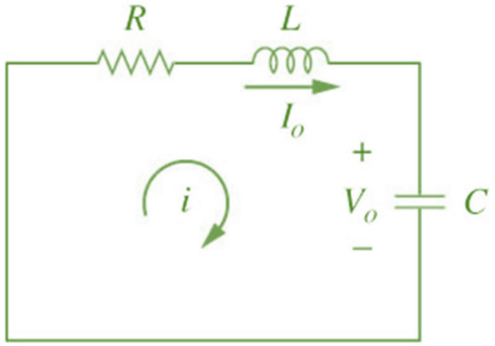


RLC Parallel

Source-Free Series RLC circuits

- The solution of the source-free series RLC circuit is called as the natural response of the circuit.
- The circuit is excited by the energy initially stored in the capacitor and inductor
- How is the second-order expression derived?

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$



- Applying KVL,

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i\,dt = 0$$

- To eliminate integral, differentiate both side with respect to time,

How to solve?

- **General second-order form**

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$

$$\Downarrow$$

$$\frac{d^2i}{dt^2} + 2\alpha\frac{di}{dt} + \omega_0^2 i = 0$$

Where $\alpha = \frac{R}{2L}$ and $\omega_0 = \sqrt{\frac{1}{LC}}$

- The types of solutions for $i(t)$depend on the relative values of $\alpha$and ω_0.
- α is called the **damping factor**, and ω_0 is called the **natural frequency** of the system.

Characteristic Equation

Let's Assume our solution is $i = Ae^{st}$

Substituting i into

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$

$$As^2e^{st} + \frac{AR}{L}se^{st} + \frac{A}{LC}e^{st} = 0$$

or

$$Ae^{st}\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right) = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Substituting with $\alpha$and ω_0:

Characteristic Equation: $s^2 + 2\alpha s + \omega_0^2 = 0$

Roots of the Characteristic Equation

Roots of the characteristic equation are:

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}, s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

OR

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

- Two roots indicate two possible solutions for current $i(t)$:

$$i_1 = A_1 e^{s_1 t}, \quad i_2 = A_2 e^{s_2 t}$$

- A total solution is a linear combination of the two solutions:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$ ← **Natural response of series RLC circuit**

where the constants A_1 and A_2 are determined from the initial values $i(0)$ and $\frac{di(0)}{dt}$.

- Note that $\frac{di(0)}{dt}$ can be found as follows:

$$Ri(0) + L\frac{di(0)}{dt} + v_c(0) = 0 \Rightarrow \frac{di(0)}{dt} = -\frac{1}{L}\left(v_c(0) + Ri(0)\right)$$

Natural Responses

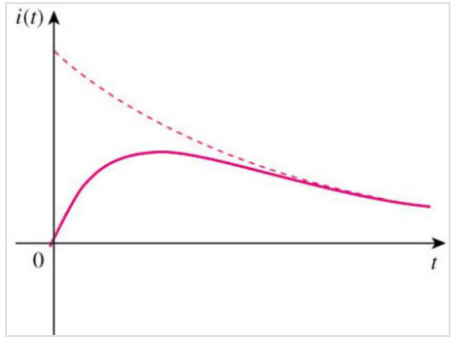
$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

- There are three types of solutions (responses) possible in a series RLC circuit, based on the following conditions:

(a) If $\alpha > \omega_0$, over-damped case

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ where } s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

s_1 and s_2 are real and negative. Response decays to zero.

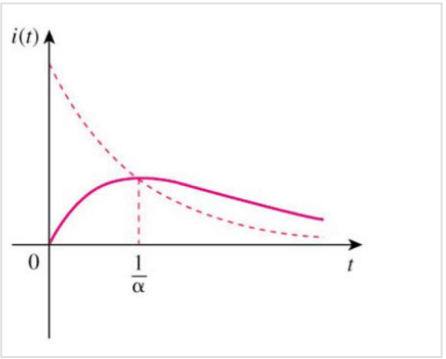


(b) If $\alpha = \omega_0$, critically damped case

$$i(t) = (A_1 + A_2 t)e^{-\alpha t} \text{ where } s_{1,2} = -\alpha$$

Two roots s_1 and s_2 are negative, real, and equal.

The response reaches a maximum value at $1/\alpha$ before decaying to zero.

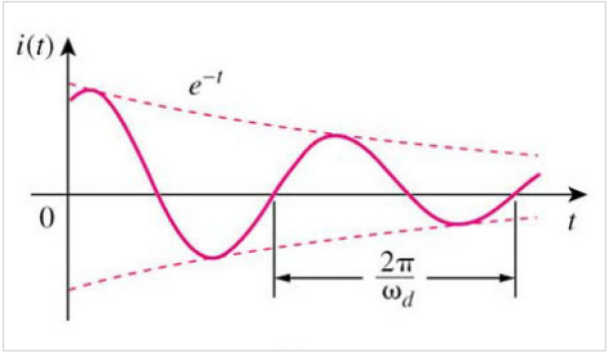


(c) If $\alpha < \omega_0$,under-damped case

$$i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) \text{ where } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$B_1 = A_1 + A_2 \text{ and } B_2 = j(A_1 - A_2)$$

Response is a decaying oscillation.



Under – damped Condition

For under-damped condition: $\alpha < \omega_0$

Roots s_1 and s_2 are complex numbers.

$$s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d \qquad s_2 = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$$

Where,

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Response is

$$i(t) = A_1 e^{(-\alpha - j\omega_d)t} + A_2 e^{(-\alpha + j\omega_d)t} = e^{-\alpha t}\left(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}\right)$$

Using Euler's identities,

$$e^{j\theta} = \cos\theta + j\sin\theta\,, e^{-j\theta} = \cos\theta - j\sin\theta$$

$$i(t) = e^{-\alpha t}[A_1(\cos\omega_d t + j\sin\omega_d t) + A_2(\cos\omega_d t - j\sin\omega_d t)]$$

$$= e^{-\alpha t}[(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t]$$

$$i(t) = e^{-\alpha t}(B_1\cos\omega_d t + B_2\sin\omega_d t)$$

where

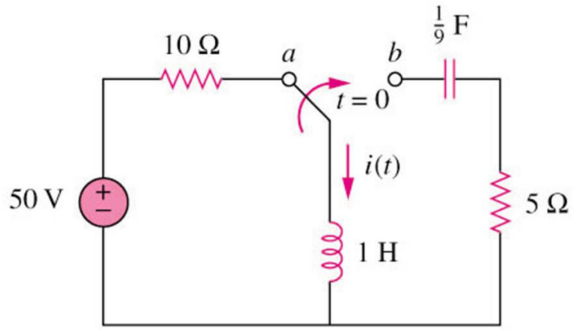
$$B_1 = A_1 + A_2, \qquad B_2 = j(A_1 - A_2)$$

Damping

- Damping in an RLC network refers to the gradual loss of initially stored energy. This damping effect is mainly due to the presence of resistance R.
- The damping factor α determines the rate at which the response decays.
- The damped oscillation observed in an underdamped response is also known as **ringing**. This oscillation occurs due to the continuous transfer of energy back and forth between the two energy storage elements, $L$and C.

Example 3

- The circuit shown below has reached steady state at $t < 0$. If the make-before-break switch moves to position b at $t = 0$, calculate $i(t)$ for $t > 0$.



Note:
$i(0^-)$ is the instant right before t =0.

$i(0^+)$ is the instant right after t =0.

Model Answer

Initial current across the inductor: $i(0^-) = \frac{50}{10} = 5\,A$; $\frac{di(0^-)}{dt} = \frac{v_L(0^-)}{L} = \frac{50 - i(0)\,R}{1} = 0$

- Initial voltage across the capacitor: $v(0^-) = 0$
- At $t = 0$, the circuit becomes a source-free RLC series circuit. $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times \frac{1}{9}}} = 3$; $\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5$

Since $\alpha < \omega_0$,we have an underdamped case.

Thus, the solution will have the form of $i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$

Where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{9 - 6.25} = 1.658$

We will determine B_1 and B_2 from the initial condition at $t = 0$. {Using KVL at $t = 0$: $L\frac{di(0)}{dt} + Ri(0) + v_C(0) = 0; \frac{di(0)}{dt} = -25$}

1 5 0

Substituting $t = 0$ in $i(t)$, $i(0) = B_1 = 5$

Differentiating $i(t)$, $\frac{di}{dt} = -\alpha e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) + \omega_d e^{-\alpha t}(-B_1 \sin \omega_d t + B_2 \cos \omega_d t)$

Substituting $t = 0$ in $\frac{di}{dt}$, $\frac{di(0)}{dt} = -\alpha B_1 + \omega_d B_2 \Rightarrow B_2 = \frac{1}{\omega_d}\left[\frac{di(0)}{dt} + \alpha B_1\right] = \frac{-25 + 2.5 \times 5}{1.658} = -7.539$

Hence $i(t) = e^{-2.5t}\,[5\cos(1.6583t) - 7.539\sin(1.6583t)]$A

Step-Response of Series RLC Circuits

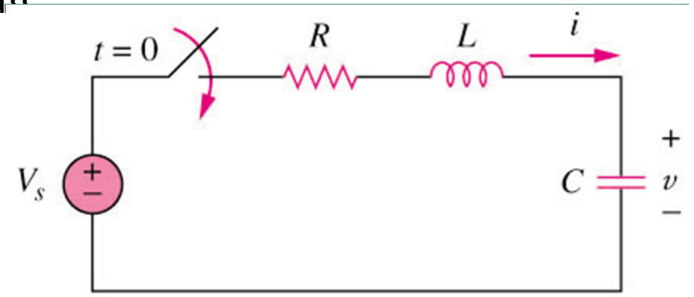
Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Step-Response of Series RLC Circuits

- The step response is obtained by the sudden application of a dc source to RLC circuit

$$L\frac{di}{dt} + Ri + v = V_s, \quad i = C\frac{dv}{dt}$$



- **The 2nd order expression**

$$\frac{d^2v(t)}{dt^2} + \frac{R}{L}\frac{dv(t)}{dt} + \frac{v(t)}{LC} = \frac{V_s}{LC}$$

- The above equation has the same form as the source-free series RLC circuit equation discussed earlier in the slides.
- The same coefficients (important in determining the frequency parameters). Different circuit variable in the equation.

The solution of the equation should have two components:

(i) the transient response v_t *and* (ii) the steady-state response $v_{ss}(t)$:

$$\boldsymbol{v(t) = v_t(t) + v_{ss}(t)}$$

- The transient response v_t is the same as that for source-free case (natural response)

$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (over-damped)

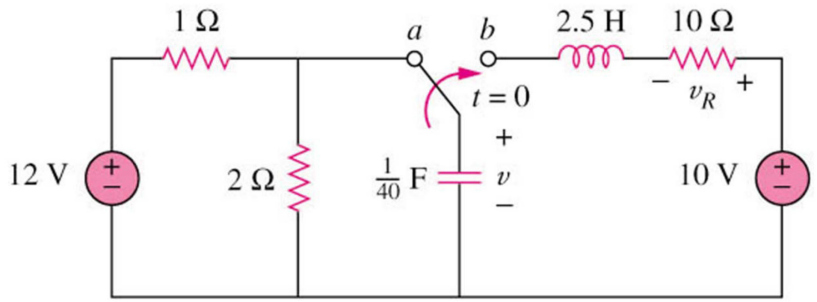
$v_t(t) = (A_1 + A_2 t)e^{-\alpha t}$ (critically damped)

$v_t(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$ (under-damped)

- The steady-state response is the final value of $v(t)$. $v_{ss}(t) = v(\infty)$
- The values of A_1, A_2, B_1 and B_2 are obtained from the initial conditions: $v(0)$ and $\frac{dv(0)}{dt}$.

Example 4

- Having been in position 'a' for a long time, the switch in the circuit below is moved to position 'b' at $t = 0$. Find $v(t)$and $v_R(t)$for $t > 0$.



Model Answer

- Initial voltage across the capacitor, $v(0) = 12 \times \frac{2}{3} = 8V$
- Initial inductor current, $i(0) = 0$
- At $t > 0$, with switch positioned at b, we have a series RLC circuit with source 10 V. Thus, the solution is

$$\boldsymbol{v(t) = v_t(t) + v_{ss}(t)}$$

$v_{ss}$is the steady-state response, and it is 10 V for the circuit. For transient response $v_t(t)$,

$$\alpha = \frac{R}{2L} = \frac{10}{2 \times 2.5} = 2$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(5/2) \times (1/40)}} = 4$$

Since $\alpha < \omega_0$,we have an **underdamped case**.

Thus the transient response will be,

$$v_t(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{16 - 4} = 3.464$$

$$v_t(t) = e^{-2t}(A_1 \cos 3.464\, t + A_2 \sin 3.464\, t)$$

- Find $A_1$and $A_2$from $v(0)$and $\frac{dv(0)}{dt}$.

$v(0)$ = 8 = 10 + A1 leads to $A_1 = -2$

The initial capacitor current is the same as the initial inductor current. $i(0) = C\left(\frac{dv(0)}{dt}\right) = 0$,therefore $\frac{dv(0)}{dt} = 0$

But, $\frac{dv}{dt} = 3.464[(-A_1 \sin 3.464\, t + A_2 \cos 3.464\, t)e^{-2t}]$
$- 2[(A_1 \cos 3.464\, t + A_2 \sin 3.464\, t)e^{-2t}]$

$\frac{dv(0)}{dt} = 0 = -2A_1 + 3.464A_2, \Rightarrow A_2 = -\frac{4}{3.464} = -1.1547$

Thus, $v(t) = \{10 + [(-2 \cos 3.464\, t - 1.1547 \sin 3.464\, t)e^{-2t}]\}\ V$

$i = C\frac{dv}{dt}, v_R = Ri = RC\frac{dv}{dt} = \frac{1}{4}\frac{dv}{dt}$

$= \frac{1}{4}[(4 - 4)\cos 3.464 + (2 \times 1.1547 + 2 \times 3.464) \sin 3.464\, t]e^{-2t}$

$$v_R(t) = \{2.31 \sin 3.464\, t\}e^{-2t}\ V$$

Source – Free Parallel RLC Circuits

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

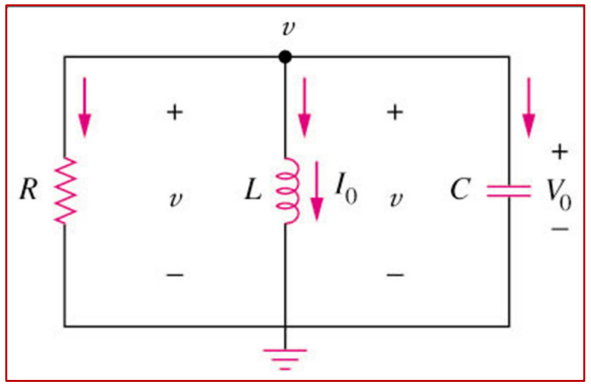
Source-Free Parallel RLC Circuits

Let $i(0) = I_0 = \frac{1}{L}\int_{-\infty}^{0} v\,(t)\,dt; \qquad v(0) = V_0$

Apply KCL to the top node: $\frac{v}{R} + \frac{1}{L}\int_{-\infty}^{t} v\ dt + C\frac{dv}{dt} = 0$

Taking the derivative with respect to t and dividing by C:

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = 0 \textbf{ (2nd order expression)}$$



Natural responses of RLC parallel circuit

There are three possible types of solutions for the following 2nd order differential equation:

$$\frac{d^2v}{dt^2} + 2\alpha\frac{dv}{dt} + \omega_0^2 v = 0 \quad \textit{where } \alpha = \frac{1}{2RC} \textit{ and } \omega_0 = \sqrt{\frac{1}{LC}}$$

1. If $\alpha > \omega_0$ **,over-damped case,** $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ *where* $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$
2. If $\alpha = \omega_0$ **,critically damped case** $v(t) = (A_2 + A_1 t)\, e^{-\alpha t}$ *where* $s_{1,2} = -\alpha$
3. If $\alpha < \omega_0$ **,under-damped case** $v(t) = e^{-\alpha t}(B_1 \cos \omega_d\, t + B_2 \sin \omega_d\, t)$ *where* $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Roots s_1, s_2 of the under-damped case are,

$$s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d; \qquad s_2 = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$$

- A1,A2 and B_1, B_2 are constants to be determined from initial conditions $v(0)$ and $\frac{dv(0)}{dt}$.

Note that $\frac{dv(0)}{dt}$ can be found easily as,

$$C\frac{dv(0^+)}{dt} + \frac{v(0^+)}{R} + i_L(0^+) = 0$$
$$\Rightarrow \frac{dv(0^+)}{dt} = \frac{-i_L(0^+)}{C}$$

Step-Response of Parallel RLC Circuits

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

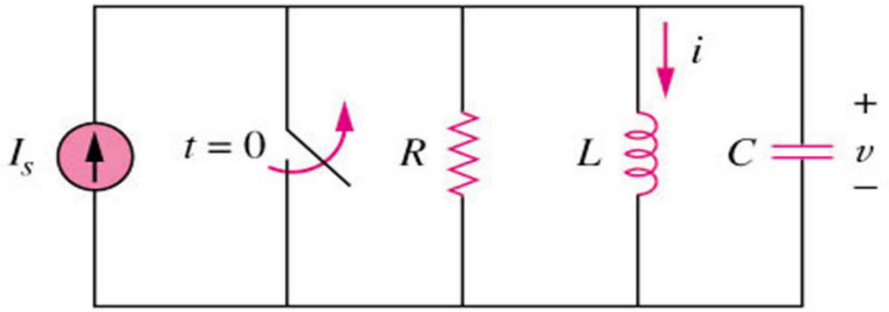
Term 3, 2026

Step-Response Parallel RLC Circuits

- The step response is obtained by the sudden application of a dc source.

$$\frac{v}{R} + i + C\frac{dv}{dt} = I_s\,, \qquad v = L\frac{di}{dt}$$

The 2nd order expression: $\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$



- It has the same form as the equation for source-free parallel RLC circuit.
- The same coefficients (important in determining the frequency parameters). Different circuit variable in the equation.

- The solution of the equation should have two components:
  the transient response $i_t(t)$ &the steady-state response $i_{ss}(t)$:

$$i(t) = i_t(t) + i_{ss}(t)$$

- The transient response $i_t$is the same as that for source-free case

$$i_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \textit{ (over–damped)}$$

$$i_t(t) = (A_1 + A_2 t)\, e^{-\alpha t} \textit{ (critical damped)}$$

$$i_t(t) = e^{-\alpha t}(A_1 \cos \omega_d\, t + A_2 \sin \omega_d\, t) \textit{ (under–damped)}$$

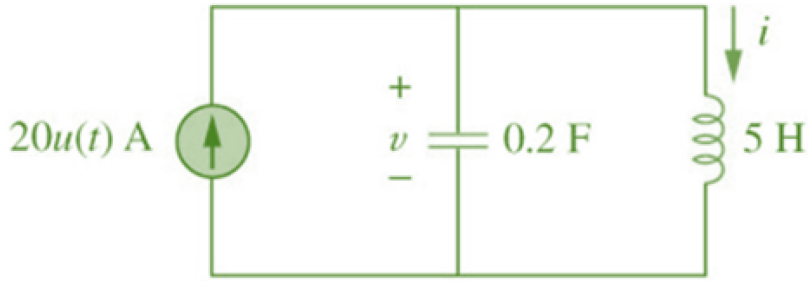
- The steady-state response is the final value of $i(t)$

$$i_{ss}(t) = i(\infty) = I_s$$

- The values of $A_1$and $A_2$are obtained from the initial conditions: $i(0)$ *and* $\frac{di(0)}{dt}$

Example 5: Source-Free Parallel RLC Circuits

- Find $i(t)$ and $v(t)$ for $t > 0$ in the circuit shown below.
  (Note: When $t < 0$, $v(0) = 0$,$i(0) = 0$;)



Model Answer

When $t < 0$, $v(0) = 0$,$i(0) = 0$; for $t > 0$, $\alpha = \frac{R}{2L} = 0$, $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2\times5}} = 1$;

$\alpha < \omega_0$, **under-damped case :** $i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$ where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1$

$i(t) = I_{ss} + (B_1 \cos t + B_2 \sin t) = 20 + B_1 \cos t + B_2 \sin t \quad \{i(\infty) = I_{ss}\}$

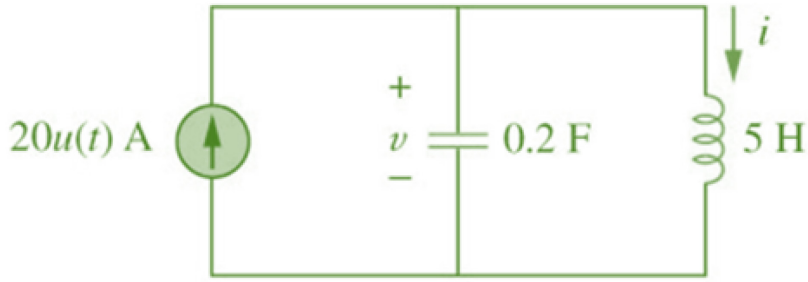
$i(0) = 0 = 20 + B_1$,therefore $B_1 = -20$ and We have $L\frac{di(0)}{dt} = v(0) = 0$

But $\frac{di(t)}{dt} = -B_1 \sin t + B_2 \cos t$; At $t = 0$, $\frac{di(0)}{dt} = 0 = 0 + B_2 \Rightarrow i(t) = 20(1 - \cos t)\ A$

$$\therefore v(t) = L\frac{di}{dt} = 5\times20 \sin t = 100 \sin t\ V$$

Example 6: Source-Free Parallel RLC Circuits

- Find $i(t)$ and $v(t)$ for $t > 0$ in the circuit shown below.
  (Note: When $t < 0$, $v(0) = 0$,$i(0) = 0$;)



Model Answer

When $t < 0$, $v(0) = 0$,$i(0) = 0$; for $t > 0$, $\alpha = \frac{R}{2L} = 0$, $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2\times5}} = 1$;

$\alpha < \omega_0$, **under-damped case :** $i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$ where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1$

$i(t) = I_{ss} + (B_1 \cos t + B_2 \sin t) = 20 + B_1 \cos t + B_2 \sin t \quad \{i(\infty) = I_{ss}\}$

$i(0) = 0 = 20 + B_1$,therefore $B_1 = -20$ and We have $L\frac{di(0)}{dt} = v(0) = 0$

But $\frac{di(t)}{dt} = -B_1 \sin t + B_2 \cos t$; At $t = 0$, $\frac{di(0)}{dt} = 0 = 0 + B_2 \Rightarrow i(t) = 20(1 - \cos t)\ A$

$$\therefore v(t) = L\frac{di}{dt} = 5\times20 \sin t = 100 \sin t\ V$$

Summary – Natural Responses

Over-damped case: $i(t)$ or $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$, where $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

Critically damped case: $i(t)$ or $v(t) = (A_2 + A_1 t)e^{-\alpha t}$, where $s_{1,2} = -\alpha$

Under damped case: $i(t)$ or $v(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$$B_1 = A_1 + A_2 \text{ and } B_2 = j(A_1 - A_2)$$

ELEC2134: Circuits and Signals

Transient analysis of 1st and 2nd systems

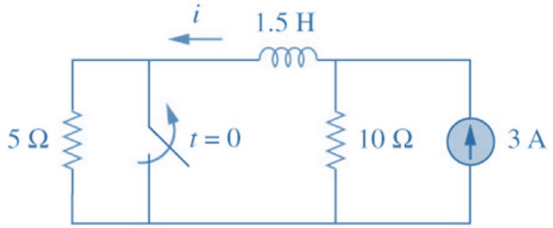
Tutorial questions

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

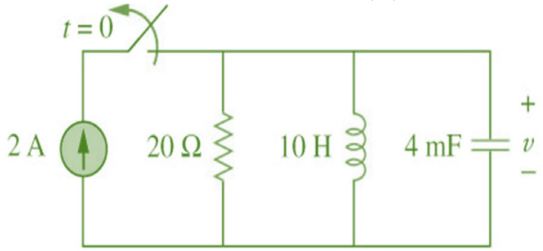
Tutorial Questions (Transient analysis of 1st and 2nd systems)

Question 1 (Pass –Level): The switch in the circuit shown below has been closed for a long time. It opens at $t = 0$. Find $i(t)$ for $t > 0$.



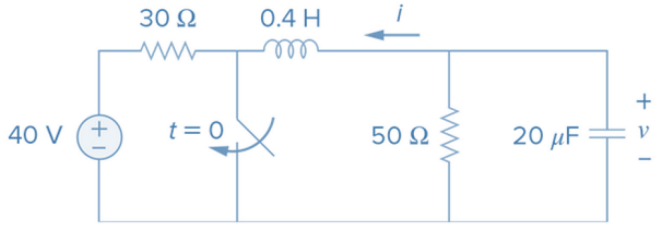
Answer: $i(t) = 2 + e^{-10t}$

Question 2 (Pass-Level): Find $v(t)$ for $t > 0$.



Answer: $v(t) = 66.67(e^{-10t} - e^{-2.5t})\ V$

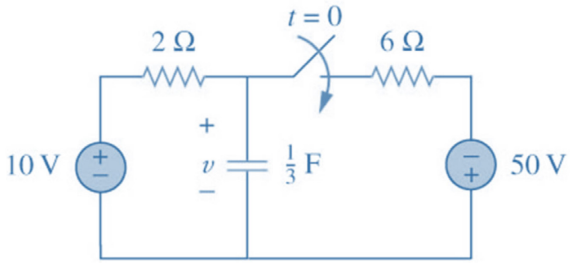
Question 3 (Pass level) : In the circuit below, find the voltage $v(t)$ for $t > 0$.



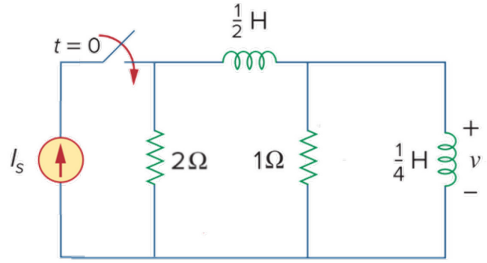
Ans: $v(t) = -5.156e^{-854t} + 30.16e^{-146t}\ V$

Question 4 (Pass - level): Find v(t) for t > 0 in the circuit below. Assume the switch has been open for a long time and is closed at t = 0. Also calculate value of v(t) at t = 0.5s.

Ans: $v(t) = 15e - 2t - 5\ ;\ v(0.5) = 0.5182$

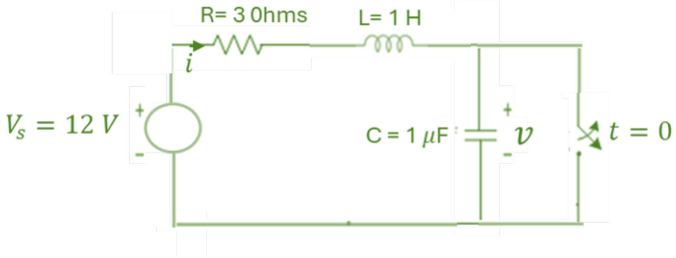


Question 5 (Higher - level): Find the voltage $v(t)$ for $t > 0$ if $I_s = 100\text{mA}$.



Ans: $v(t) = \frac{1}{15}(e^{-2t} - e^{-8t})$

Question 6 (Higher Level): Find In the circuit below, find the voltage $v(t)$ across the capacitor for $t > 0$ and sketch $v(t)$ vs t.



Ans: $v(t) = e^{-\frac{3}{2}t}(-12\cos(10^3 t) + 4.10^3 \sin(10^3 t)) + 12$